

Confidence Intervals II

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Stat 141 | Fall 2026

Monday, Week 9

Reminders/Announcements

- sUms colloquium today
- Course assistants would love to see more of you in office hours!
- I'll be a few minutes late to office hours today

Goals for today

- Review the concept of a 95% confidence interval
- Discuss (one way) of creating confidence intervals with different confidence levels.
- Interpret confidence intervals and discuss common misconceptions

Review

Setting

- There is some **population** we're interested in studying.
  - e.g., Reed College students
- There is a population **parameter** we want to know
  - e.g., average nightly hours of sleep (μ)
- We draw a **sample** from the population with sample size n
  - e.g., We survey $n = 50$ Reed students
- We provide a **point estimate** of the parameter with a statistic
  - e.g., average hours of sleep in our sample ($\bar{x} = 8.02$ hours)
- We construct an **interval estimate** centered at our statistic
  - e.g., 8.02 ± 0.16 hours

Warm-Up

1. What are the differences between a sampling distribution and a bootstrap distribution?
2. Suppose we created two confidence intervals based on two **distinct** samples: one of size $n = 10$ and another of size $n = 100$. How might they differ?

02:00

Confidence Intervals

- A **confidence interval** gives a range of plausible values for a parameter. It usually takes the form:

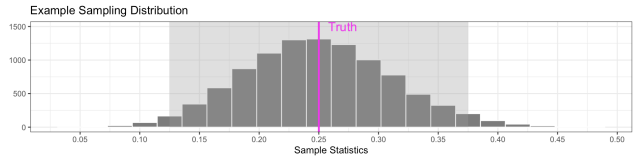
$$\text{Statistic} \pm \text{Margin of Error (ME)}$$

- The Margin of Error (ME) is partially determined by our sample size:
  - Bigger samples yield smaller ME and interval (more data $\implies$ more certainty)
  - Smaller samples yield wider ME and interval (less data $\implies$ less certainty)
- Every confidence interval has a **confidence level:** the percentage of samples that would yield a confidence interval containing the true parameter value.
- For example, were we to:
  - Repeatedly draw samples from the population
  - Create 95% confidence intervals in each sample
  - 95% of those confidence intervals would contain the true parameter

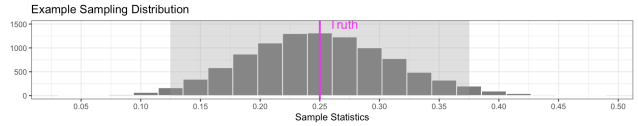
Sampling Distributions can determine the Margin of Error

- Reminders:
  - Sampling distribution is *often* bell-shaped, with the mean equal to the parameter
  - In bell-shaped (normal) distributions, 95% of observations lie in the range:

$$\text{Mean} \pm 2 * \text{Standard Error}$$



Sampling Distributions can determine the Margin of Error



Implication: in the sampling distribution, 95% of sample statistics lie in the range:

$$\text{Parameter} \pm 2 * \text{Standard Error}$$

where the **Standard Error** is the standard deviation of the sampling distribution

- So, 95% of sample statistics are within $2 * \text{SE}$ from the parameter!
- Thus, 95% confidence intervals usually take the form:

$$\text{Statistic} \pm 2 * \text{Standard Error (SE)}$$

- We typically estimate the SE (i) with the bootstrap, or (ii) with a formula (we'll see this in a few weeks)

Review: Bootstrapping Confidence Intervals

Example: Reproduction Rate for Covid-19

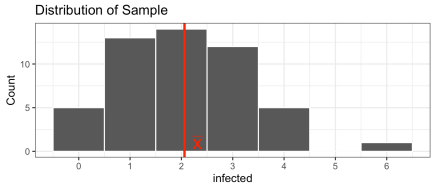
Researchers are interested in the COVID-19 **reproduction rate** (the average number of individuals each infected person further infects). They sample 50 infected individuals and perform contact tracing.

```
1 covid %>% count(infected)
```

```
  infected  n
1        0  5
2        1 13
3        2 14
4        3 12
5        4  5
6        6  1
```

```
1 covid %>% summarize(mean_infected = mean(infected
```

```
  mean_infected
1          2.06
```



- **Goal:** Create an interval of plausible values for the reproduction rate.
- **Q:** What is the population? What is the parameter?
- **Q:** What is the sample? What is the statistic?

Bootstrap Reproduction Rate

We can use our sample to create a 95% confidence interval. **What is each step doing, and why?**

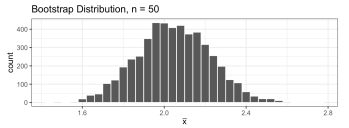
1. Step 1:

```
1 set.seed(121)
2 bootstrap_samples <- covid %>% rep_sample_n(size = 50, replace = TRUE, reps = 5000)
```

2. Step 2:

```
1 bootstrap_stats <- bootstrap_samples %>% group_by(replicate) %>% summarize(x_bar = mean(infected))
```

3. Step 3:



4. Step 4:

```
1 bootstrap_stats %>% summarize(SE = sd(x_bar))
```

```
# A tibble: 1 × 1
     SE
  <dbl>
1 0.181
```

5. Step 5:

$$\bar{x} \pm 2 \cdot SE \implies 2.06 \pm 2 \cdot 0.181$$

03:00

Example: Reproduction Rate for Covid-19

Our 95% confidence interval for the reproduction rate is:

$$\bar{x} \pm 2 \cdot SE \implies 2.06 \pm 2 \cdot 0.181$$

$$= [1.698, 2.422].$$

Generalized Confidence Intervals

Generalized Confidence Intervals

- In the previous example, we used our knowledge that for approximately bell-shaped sampling distributions, 95% of sample statistics are within 2 SE of the population parameter
  - Suppose we instead want a **different success rate** for our estimation method
  - Or suppose we want interval estimates for **sampling distributions that are NOT bell-shaped**

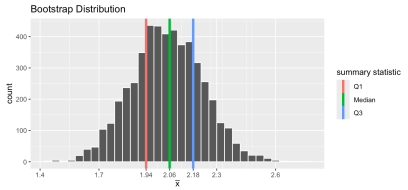
General Confidence Intervals

The $C\%$ confidence interval for a parameter is an interval estimate that is computed from sample data by a method that captures the parameter for $C\%$ of all samples.

- We can achieve this using by again using the bootstrap approximation to the sampling distribution

Review: Percentiles and Quantiles

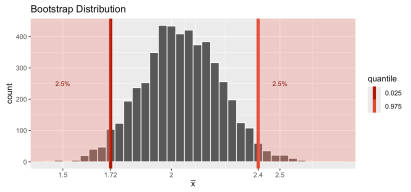
- For a number k between 0 and 100, the kth **percentile** of a distribution is the value so that $k\%$ of the data is less than or equal to that value.
  - The median is the 50th percentile of a distribution
  - 1st/3rd quartiles (Q1/Q3) are the 25th and 75th percentiles, respectively.



- For a number p between 0 and 1, the p **quantile** of a distribution is the value so that a proportion p of the data is less than or equal to that value.
  - The median is the 0.5 quantile of a distribution
  - 1st/3rd quartiles (Q1/Q3) are the 0.25 and 0.75 quantiles, respectively.

Quantiles and Percentiles

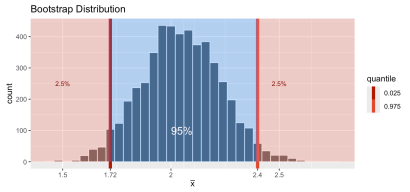
By definition, 2.5% of the data is less than the .025 quantile, and 2.5% of the data is greater than the .975 quantile



- This means that 95% of the data is between the .025 and the .975 quantiles.

Quantiles and Percentiles

By definition, 2.5% of the data is less than the .025 quantile, and 2.5% of the data is greater than the .975 quantile



- This means that 95% of the data is between the .025 and the .975 quantiles.
- *For sampling distributions that are bell-shaped*, the .025 quantile is about $2 \cdot SE$ below the mean, and the .975 quantile is about $2 \cdot SE$ above the mean
- So using the .025 and .975 quantiles is roughly equivalent to forming a 95% CI as: Statistic $\pm$ 2 $*$ SE!

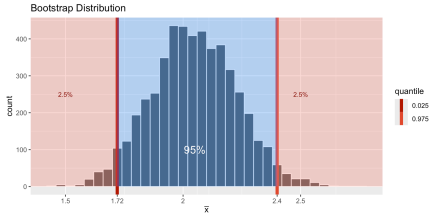
95% Confidence Interval: 2 ways

1. Statistic $\pm$ 2 $*$ SE

- Statistic $(\bar{x}) = 2.06$
- SE $= 0.18$
- 95% CI $= 1.70$ to 2.42

- **Requires a bell-shaped sampling distribution**

2. Percentile Method




```
1 quantile(bootstrap_stats$x_bar, c(0.025, 0.975))
2.5% 97.5%
1.72  2.40
```

- **Sampling distribution does not need to be bell-shaped**

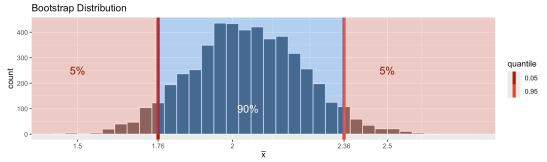
Percentile Method for Confidence Intervals

- **Percentile Method:** For a $C\%$ Confidence Interval, report the quantiles of the *bootstrap distribution* such that:
  - $C\%$ lies in the middle
  - $\frac{(100-C)}{2}\%$ lies on either end (i.e., "the rest" is evenly distributed on the ends)

- **Ex:** For $C\% = 95\%$ (i.e., a 95% Confidence Interval), we want
  - 95% in the middle
  - $\frac{(100-C)}{2}\% = \frac{(100-95)}{2}\% = 2.5\%$ **on either end**
  - Report the **0.025 quantile** (or 2.5th percentile) and the **0.975 quantile** (or 97.5th percentile) as our confidence intervals

The Percentile Method: Example

- Suppose we want to construct a **90% confidence interval** for the reproduction rate
  - Find the .05 and .95 quantiles in the bootstrap distribution.
  - 90% of bootstrap sample statistics will be between these values



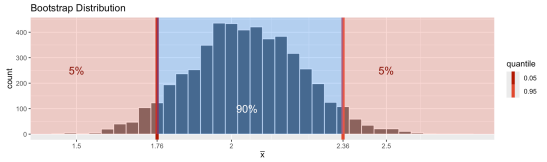
- We can use the `quantile` function in R to calculate the .05 and .95 quantiles

```
quantile(bootstrap_stats$x_bar, c(.05, .95))
```

```
  5%  95%
1.76 2.36
```


The Percentile Method: Example

- Suppose we want to construct a **90% confidence interval** for the reproduction rate
  - Find the .05 and .95 quantiles in the bootstrap distribution.
  - 90% of bootstrap sample statistics will be between these values



- Our 90% confidence interval is therefore 1.76 to 2.36

```
quantile(bootstrap_stats$x_bar, c(.05, .95))
```

```
  5%  95%
1.76 2.36
```


Percentile Method: Practice

With neighbor(s), name the quantiles of the bootstrap distribution you would need for:

1. 80% Confidence Interval
2. 99% Confidence Interval
3. 2% Confidence Interval *(You would never do this! This is just for fun (:)*

01:30

Width of Confidence Intervals

Two factors determine the width of a confidence interval:

1. **Sample Size**
   - The Standard Error of the sampling distribution decreases as sample size increases.
   - Smaller sample size $\implies$ larger interval
   - Larger sample size $\implies$ smaller interval
2. **Confidence Level**
   - Decreasing the confidence level brings the relevant quantiles closer to the middle, decreasing the width of the interval.
   - Higher confidence level $\implies$ larger interval
   - Lower confidence level $\implies$ smaller interval

Width of Confidence Intervals

Confidence Intervals get smaller with:

- a larger sample size
- a lower confidence level

Note: These reasons for getting smaller are competing in terms of certainty!

- With a larger sample size, the interval gets smaller because we're *more* certain the statistic is close to the parameter.
- With a lower confidence level, we become *less* certain the interval will contain the true parameter.

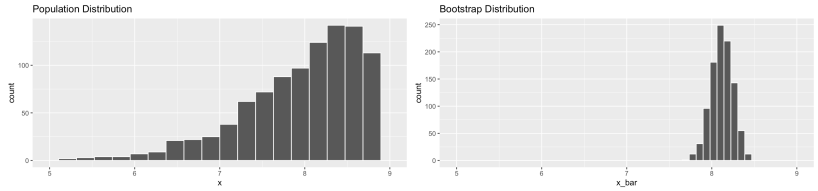
> **Reminder:** While a lower confidence level gives you a smaller interval, there is a cost! (i.e., lower success rate)

Confidence Interval Misunderstandings

Misunderstanding 1

Suppose we wish to estimate the number of hours a Reed student sleeps on a typical night. We obtain the following 95% confidence interval: $(7.86, 8.34)$

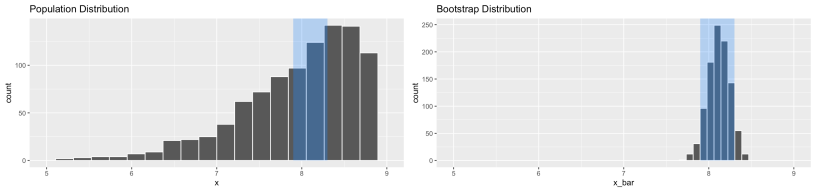
1. A 95% confidence interval **does not** contain 95% of observations in the *population.*



Misunderstanding 1

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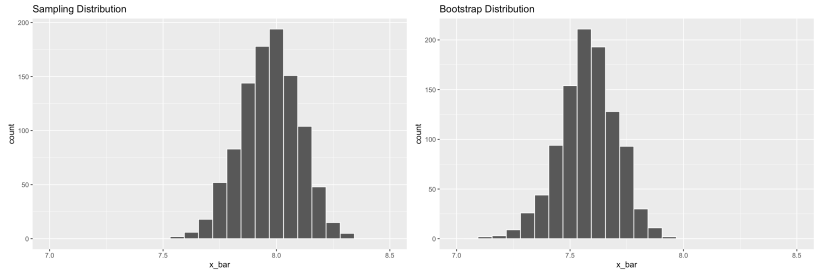
1. A 95% confidence interval **does not** contain 95% of observations in the *population.*



- Saying that 95% of all Reed students sleep between 7.86 and 8.34 hours should just feel wrong. That's a pretty narrow interval!

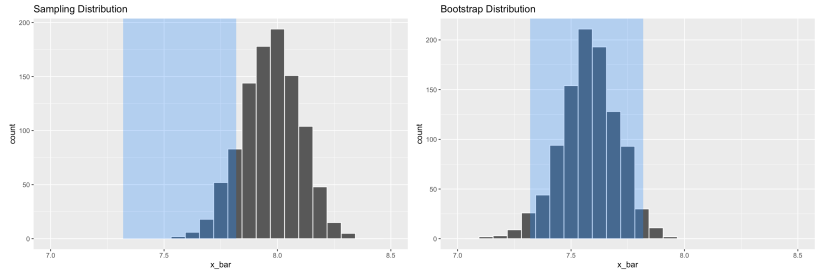
Misunderstanding 2

2. A 95% confidence interval **does not** mean that 95% of all sample means fall within the given range.



Misunderstanding 2

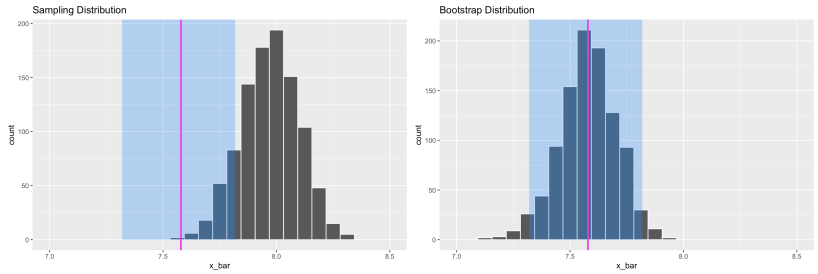
2. A 95% confidence interval **does not** mean that 95% of all sample means fall within the given range.



- **Q:** Why do the sampling distribution and bootstrap distribution look different?

Misunderstanding 2

2. A 95% confidence interval **does not** mean that 95% of all sample means fall within the given range.



- **Q:** Why do the sampling distribution and bootstrap distribution look different?

Misunderstanding 3

3. Given a 95% confidence interval, **Do Not Say:** "There is a 95% chance that the true parameter falls within my interval."

- Once we take a sample and calculate a confidence interval, there's no more randomness!
  - The interval either does or doesn't contain the (unknown) parameter.
- This is may seem like arguing over semantics – but it's an important distinction!

Instead, say either:

- "If we were to take many samples and calculate a 95% confidence interval for each, then 95% of them would contain the true parameter"
- "We are 95% confident that the true parameter is in our confidence interval"

Recap

- Two ways to construct 95% confidence intervals *specifically*
  - **SE Method**: $\text{Statistic} \pm 2 * \text{SE}$ (estimate SE via bootstrap)
  - **Percentile Method**: take the 0.025 and 0.975 quantiles of the estimated sampling distribution (via bootstrap)
- If we want a confidence level that *isn't* 95%, we can easily adapt the percentile method by taking different quantiles from our bootstrap distribution.
  - We can actually generalize the SE method to other levels too, but we won't see that for a few weeks.
- We should always interpret confidence intervals carefully, and in context.

Next time

- Hypothesis testing!